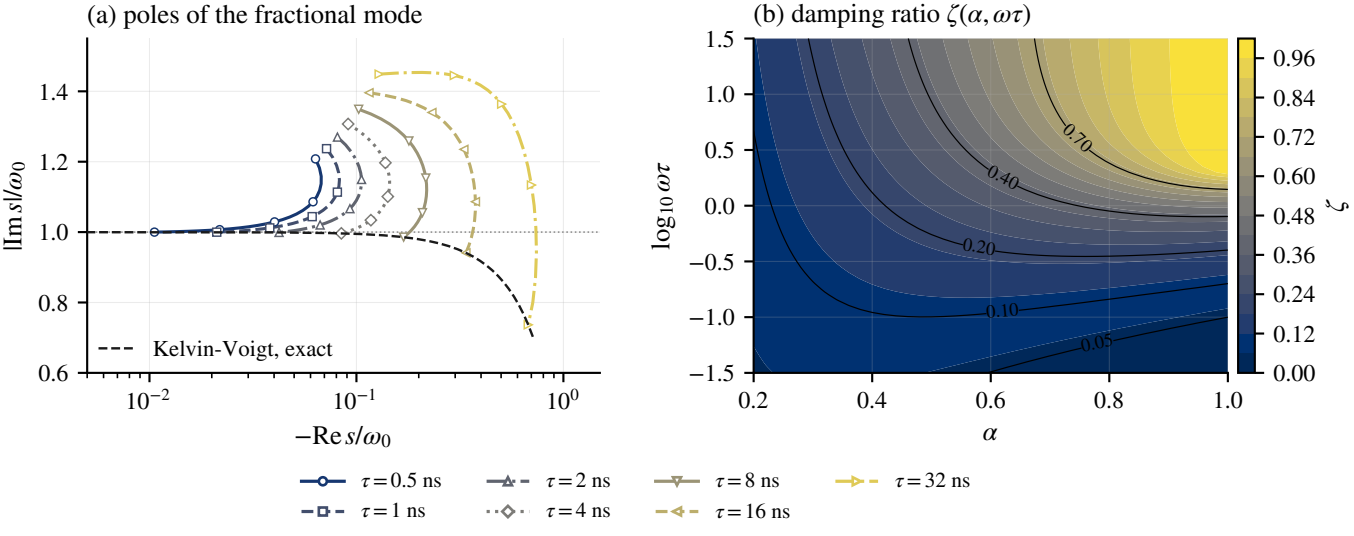
(a) poles of the fractional mode
(b) damping ratio $\zeta(\alpha, \omega\tau)$
$\mathrm{Im}\, s/\omega_0$
$-\mathrm{Re}\, s/\omega_0$
Kelvin-Voigt, exact
$\log_{10} \omega\tau$
α
ζ
0.70
0.40
0.20
0.10
0.05
$\tau = 0.5$ ns
$\tau = 1$ ns
$\tau = 2$ ns
$\tau = 4$ ns
$\tau = 8$ ns
$\tau = 16$ ns
$\tau = 32$ ns